

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12398734
Kind Code	B2
Date of Patent	August 26, 2025
Inventor(s)	Le Doux, Jr.; Norman R. et al.

Pistons for use in fluid exchange devices and related devices, systems, and methods

Abstract

Pistons and related methods may be configured to separate fluids and to at least partially prohibit one fluid from traveling to one side of the piston from another side of the piston. Pressure exchange devices and systems may include such pistons.

Inventors: Le Doux, Jr.; Norman R. (Orefield, PA), Bandi; Jason (Irving, TX), Havrilla; Neil (Coplay, PA), Olexson; David (Irving, TX), Dreiss; Andreas (Hamburg, DE)

Applicant: FLOWSERVE PTE. LTD. (Singapore, SG)

Family ID: 1000008778796

Assignee: Flowserve Pte. Ltd. (Singapore, SG)

Appl. No.: 18/545378

Filed: December 19, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20240117820 A1	Apr. 11, 2024

Related U.S. Application Data

continuation parent-doc US 17671283 20220214 US 11852169 child-doc US 18545378
continuation parent-doc US 16679037 20191108 US 11286958 child-doc US 17671283
us-provisional-application US 62758373 20181109

Publication Classification

Int. Cl.: F04F13/00 (20090101); **F16K11/07** (20060101); **F16K31/08** (20060101); F15B3/00 (20060101)

U.S. Cl.:

CPC F04F13/00 (20130101); **F16K11/0716** (20130101); **F16K31/084** (20130101); F15B3/00 (20130101)

Field of Classification Search

CPC: F04F (13/00)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
1577242	12/1925	Andersen	N/A	N/A
1647189	12/1926	Mueller	N/A	N/A
1647734	12/1926	Kelly	N/A	N/A
1769672	12/1929	Blair	N/A	N/A
1970591	12/1933	Barks	N/A	N/A
2365046	12/1943	Bottomley	N/A	N/A
2600836	12/1951	Boyd	N/A	N/A
2615465	12/1951	Woodward	N/A	N/A
2881706	12/1958	Haynes	N/A	N/A
3089504	12/1962	Crawford	N/A	N/A
3223173	12/1964	Paul, Jr.	N/A	N/A
3347554	12/1966	Jagger	N/A	N/A
3570510	12/1970	Tsutsumi	N/A	N/A
3583606	12/1970	Ewald	N/A	N/A
3595265	12/1970	Cryder	N/A	N/A
3612361	12/1970	Ewald	N/A	N/A
3661167	12/1971	Hussey	N/A	N/A
3661400	12/1971	Weinand	N/A	N/A
3675825	12/1971	Morane	N/A	N/A
3675935	12/1971	Ludwig	N/A	N/A
3741243	12/1972	Deibler	N/A	N/A
3749291	12/1972	Prussin	N/A	N/A
3756273	12/1972	Hengesbach	N/A	N/A
3776278	12/1972	Allen	N/A	N/A
4024891	12/1976	Engel	N/A	N/A
4123332	12/1977	Rotter	N/A	N/A
4133346	12/1978	Smith	N/A	N/A
4134454	12/1978	Taylor	N/A	N/A
4176063	12/1978	Tyler	N/A	N/A
4234010	12/1979	Jenkins	N/A	N/A
4236547	12/1979	Harasewych	N/A	N/A
4244555	12/1980	Maggioni	N/A	N/A
4249868	12/1980	Kotyk	N/A	N/A

4308103	12/1980	Rotter	N/A	N/A
4321021	12/1981	Pauliukonis	N/A	N/A
4350176	12/1981	Lace	N/A	N/A
4412632	12/1982	Berger	N/A	N/A
4424917	12/1983	Berger	N/A	N/A
4479356	12/1983	Gill	N/A	N/A
4510963	12/1984	Presley	N/A	N/A
4518006	12/1984	Hoffmann	N/A	N/A
4566485	12/1985	Ruhle	N/A	N/A
4570853	12/1985	Schmied	N/A	N/A
4579511	12/1985	Burns	N/A	N/A
4586692	12/1985	Stephens	N/A	N/A
4627461	12/1985	Gordon	N/A	N/A
4726530	12/1987	Miller	N/A	N/A
4768542	12/1987	Morris	N/A	N/A
4834193	12/1988	Leitko, Jr.	N/A	N/A
4999872	12/1990	Jentsch	N/A	N/A
5033557	12/1990	Askew	N/A	N/A
5070817	12/1990	Momont	N/A	N/A
5172918	12/1991	Pecht	N/A	N/A
5232013	12/1992	Morris	N/A	N/A
5234031	12/1992	Pickett	N/A	N/A
5240036	12/1992	Morris	N/A	N/A
5299859	12/1993	Tackett	N/A	N/A
5300041	12/1993	Haber	N/A	N/A
5357995	12/1993	King	N/A	N/A
5797429	12/1997	Shumway	N/A	N/A
5951169	12/1998	Oklejas	N/A	N/A
5992289	12/1998	George	N/A	N/A
6036435	12/1999	Oklejas	N/A	N/A
6126418	12/1999	Sinnl	N/A	N/A
6293245	12/2000	Bock	N/A	N/A
RE37921	12/2001	Martin	N/A	N/A
6516897	12/2002	Thompson	N/A	N/A
6540487	12/2002	Polizos	N/A	N/A
6607368	12/2002	Ross	N/A	N/A
6647938	12/2002	Gaessler	N/A	N/A
6659731	12/2002	Hauge	N/A	N/A
7128084	12/2005	Long	N/A	N/A
7201557	12/2006	Stover	N/A	N/A
7306437	12/2006	Hauge	N/A	N/A
7474013	12/2008	Greenspan	N/A	N/A
7670482	12/2009	Wietham	N/A	N/A
7871522	12/2010	Stover	N/A	N/A
RE42432	12/2010	Stover	N/A	N/A
7997853	12/2010	Pique	N/A	N/A
8075281	12/2010	Martin	N/A	N/A
8297303	12/2011	DeSantis	N/A	N/A
8360250	12/2012	Nguyen	N/A	N/A
8465000	12/2012	Bartell, Jr.	N/A	N/A

8579603	12/2012	Oklejas	N/A	N/A
8603218	12/2012	Montie	N/A	N/A
8622714	12/2013	Andrews	N/A	N/A
9108162	12/2014	Takahashi	N/A	N/A
9163737	12/2014	Andersson	N/A	N/A
9328743	12/2015	Hirosawa	N/A	N/A
9435354	12/2015	Lehner	N/A	N/A
9440895	12/2015	Arluck	N/A	N/A
9500394	12/2015	Manzo	N/A	N/A
9523261	12/2015	Flores	N/A	N/A
9546671	12/2016	Hirosawa	N/A	N/A
9556736	12/2016	Sigurdsson	N/A	N/A
9587752	12/2016	Montague	N/A	N/A
9604889	12/2016	Arluck	N/A	N/A
9611948	12/2016	Andersson	N/A	N/A
9683574	12/2016	Winkler	N/A	N/A
9695795	12/2016	Martin	N/A	N/A
9739128	12/2016	Ghasripoor	N/A	N/A
9739275	12/2016	Robison	N/A	N/A
9759054	12/2016	Gay	N/A	N/A
9764272	12/2016	Martin	N/A	N/A
9835018	12/2016	Krish	N/A	N/A
9885372	12/2017	Arluck	N/A	N/A
9920774	12/2017	Ghasripoor	N/A	N/A
9945210	12/2017	Theodossiou	N/A	N/A
9945216	12/2017	Ghasripoor	N/A	N/A
9970281	12/2017	Ghasripoor	N/A	N/A
9975789	12/2017	Ghasripoor	N/A	N/A
9976573	12/2017	Martin	N/A	N/A
10001030	12/2017	Krish	N/A	N/A
10006524	12/2017	Crump	N/A	N/A
10024496	12/2017	Hauge	N/A	N/A
10030372	12/2017	Di Monte, Sr.	N/A	N/A
10072675	12/2017	McLean, Jr.	N/A	N/A
10119379	12/2017	Richter	N/A	N/A
10125796	12/2017	Hauge	N/A	N/A
10138907	12/2017	Pinto	N/A	N/A
10167710	12/2018	Ghasripoor	N/A	N/A
10167712	12/2018	Ghasripoor	N/A	N/A
2002/0025264	12/2001	Polizos	N/A	N/A
2002/0092571	12/2001	Misumi	N/A	N/A
2004/0118462	12/2003	Baumann	N/A	N/A
2005/0103386	12/2004	Magda	N/A	N/A
2005/0166978	12/2004	Brueckmann	N/A	N/A
2006/0145426	12/2005	Schroeder	N/A	N/A
2006/0196474	12/2005	Magel	N/A	N/A
2006/0231577	12/2005	Powling	N/A	N/A
2007/0204916	12/2006	Clayton	N/A	N/A
2009/0057084	12/2008	Mahawili	N/A	N/A
2009/0104046	12/2008	Martin	N/A	N/A

2009/0313737	12/2008	Richard	N/A	N/A
2012/0024249	12/2011	Fuhrmann	N/A	N/A
2012/0061595	12/2011	Canjuga	N/A	N/A
2012/0067825	12/2011	Pique	N/A	N/A
2014/0026608	12/2013	Manzo	N/A	N/A
2014/0048143	12/2013	Lehner	N/A	N/A
2014/0284058	12/2013	Watson	N/A	N/A
2015/0096739	12/2014	Ghasripoor	N/A	N/A
2015/0130142	12/2014	Zheng	N/A	N/A
2015/0184540	12/2014	Winkler	N/A	N/A
2015/0184678	12/2014	Arluck	N/A	N/A
2015/0292310	12/2014	Ghasripoor	N/A	N/A
2016/0032691	12/2015	Richter	N/A	N/A
2016/0032702	12/2015	Gay	N/A	N/A
2016/0039054	12/2015	Ghasripoor	N/A	N/A
2016/0062370	12/2015	Gaines-Germain	N/A	N/A
2016/0101307	12/2015	Montague	N/A	N/A
2016/0102536	12/2015	Knoeller	N/A	N/A
2016/0138649	12/2015	Anderson	N/A	N/A
2016/0146229	12/2015	Martin	N/A	N/A
2016/0153551	12/2015	Schiele	N/A	N/A
2016/0160849	12/2015	Gains-Germain	N/A	N/A
2016/0160881	12/2015	Anderson	N/A	N/A
2016/0160882	12/2015	Morphew	N/A	N/A
2016/0160887	12/2015	Anderson	N/A	N/A
2016/0160888	12/2015	Morphew	N/A	N/A
2016/0160889	12/2015	Hoffman	N/A	N/A
2016/0160890	12/2015	Anderson	N/A	N/A
2016/0160917	12/2015	Deshpande	N/A	N/A
2016/0222985	12/2015	Oklejas, Jr.	N/A	F04B 9/08
2016/0377096	12/2015	Lehner	N/A	N/A
2017/0108131	12/2016	Andersson	N/A	N/A
2017/0130743	12/2016	Anderson	N/A	N/A
2017/0254474	12/2016	Sauer	N/A	N/A
2017/0306986	12/2016	McLean, Jr.	N/A	N/A
2017/0306987	12/2016	Theodossiou	N/A	N/A
2017/0350428	12/2016	Martin	N/A	N/A
2017/0370500	12/2016	Haines	N/A	N/A
2018/0030968	12/2017	Verma	N/A	N/A
2018/0056211	12/2017	Seabrook	N/A	N/A
2018/0087364	12/2017	Krish	N/A	N/A
2018/0094648	12/2017	Hoffman	N/A	N/A
2018/0120197	12/2017	Di Monte	N/A	N/A
2018/0195370	12/2017	Theodossiou	N/A	N/A
2018/0209254	12/2017	Ghasripoor	N/A	N/A
2018/0252239	12/2017	Martin	N/A	N/A
2018/0306672	12/2017	Pattom	N/A	N/A
2018/0347601	12/2017	Hoffman	N/A	N/A
2019/0071340	12/2018	Imrie	N/A	N/A
2020/0149362	12/2019	Terwilliger	N/A	N/A

2020/0149380	12/2019	Procita	N/A	N/A
2020/0149556	12/2019	Judge	N/A	N/A
2020/0149557	12/2019	Le Doux, Jr.	N/A	N/A
2020/0149657	12/2019	Christian	N/A	N/A
2020/0150698	12/2019	Judge	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
101705930	12/2009	CN	N/A
102421513	12/2011	CN	N/A
103114924	12/2012	CN	N/A
103298731	12/2012	CN	N/A
105683586	12/2015	CN	N/A
206158951	12/2016	CN	N/A
0163897	12/1987	EP	N/A
1486706	12/2003	EP	N/A
1948942	12/2007	EP	N/A
3177429	12/2016	EP	N/A
6386657	12/2017	JP	N/A
20010081640	12/2000	KR	N/A
503937	12/2001	NZ	N/A
2284423	12/2005	RU	N/A
2010145595	12/2011	RU	N/A
2511638	12/2013	RU	N/A
410199	12/1973	SU	N/A
2002066816	12/2001	WO	N/A
2009113877	12/2008	WO	N/A
2010031162	12/2009	WO	N/A
2016022706	12/2015	WO	N/A
2016063194	12/2015	WO	N/A
2017083500	12/2016	WO	N/A
2018035201	12/2017	WO	N/A
2018085740	12/2017	WO	N/A

OTHER PUBLICATIONS

PCT International Patent Application No. PCT/US2019/060621, International Search Report dated Mar. 19, 2020, 2 pp. cited by applicant

PCT International Patent Application No. PCT/US2019/060621, Written Opinion dated Mar. 19, 2020, 5 pp. cited by applicant

Vorteq Pure Grit, This changes everything, Brochure, Energy Recovery Inc, 8 pages, Dec. 2014. cited by applicant

Primary Examiner: Rinehart; Kenneth

Assistant Examiner: Gardner; Nicole

Attorney, Agent or Firm: BK RIP LLC

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) This application is a continuation of U.S. patent application Ser. No. 17/671,283, filed Feb. 14, 2022, for “PISTONS FOR USE IN FLUID EXCHANGE DEVICES AND RELATED DEVICES, SYSTEMS, AND METHODS,” which is a continuation of U.S. patent application Ser. No. 16/679,037, filed Nov. 8, 2019, for “PISTONS FOR USE IN FLUID EXCHANGE DEVICES AND RELATED DEVICES, SYSTEMS, AND METHODS,” which application claims the benefit under 35 U.S.C. § 119(e) of U.S. Provisional Patent Application Ser. No. 62/758,373, filed Nov. 9, 2018, for “PISTONS FOR USE IN FLUID EXCHANGE DEVICES AND RELATED DEVICES, SYSTEMS, AND METHODS,” the disclosure of each of which is hereby incorporated herein in its entirety by this reference.

TECHNICAL FIELD

(1) The present disclosure relates generally to pistons for exchange devices. More particularly, embodiments of the present disclosure relate to pistons for use in fluid exchange devices for one or more of exchanging properties (e.g., pressure) between fluids and systems and methods.

BACKGROUND

(2) Industrial processes often involve hydraulic systems including pumps, valves, impellers, etc. Pumps, valves, and impellers may be used to control the flow of the fluids used in the hydraulic processes. For example, some pumps may be used to increase (e.g., boost) the pressure in the hydraulic system, other pumps may be used to move the fluids from one location to another. Some hydraulic systems include valves to control where a fluid flows. Valves may include control valves, ball valves, gate valves, globe valves, check valves, isolation valves, combinations thereof, etc.

(3) Some industrial processes involve the use of caustic fluids, abrasive fluids, and/or acidic fluids. These types of fluids may increase the amount of wear on the components of a hydraulic system. The increased wear may result in increased maintenance and repair costs or require the early replacement of equipment. For example, abrasive, caustic, or acidic fluid may increase the wear on the internal components of a pump such as an impeller, shaft, vanes, nozzles, etc. Some pumps are rebuildable and an operation may choose to rebuild a worn pump replacing the worn parts which may result in extended periods of downtime for the worn pump resulting in either the need for redundant pumps or a drop in productivity. Other operations may replace worn pumps at a larger expense but a reduced amount of downtime.

(4) Well completion operations in the oil and gas industry often involve hydraulic fracturing (often referred to as fracking or fracing) to increase the release of oil and gas in rock formations. Hydraulic fracturing involves pumping a fluid (e.g., frac fluid, fracking fluid, etc.) containing a combination of water, chemicals, and proppant (e.g., sand, ceramics) into a well at high pressures. The high pressures of the fluid increases crack size and crack propagation through the rock formation releasing more oil and gas, while the proppant prevents the cracks from closing once the fluid is depressurized. Fracturing operations use high-pressure pumps to increase the pressure of the fracking fluid. However, the proppant in the fracking fluid increases wear and maintenance on and substantially reduces the operation lifespan of the high-pressure pumps due to its abrasive nature.

BRIEF SUMMARY

(5) Various embodiments may include a device or system for exchanging pressure between fluids. The device may include at least one tank, at least one piston in the at least one tank, and a valve device. The tank may include a clean side for receiving a clean fluid at a higher pressure and a dirty side for receiving a downhole fluid (e.g., a fracking fluid, a drilling fluid) at a lower pressure. The piston may be configured to separate the clean fluid from the downhole fluid to at least partially prohibit fluid the downhole fluid from traveling from the dirty side to the clean side. The valve device may be configured to selectively place the clean fluid at the higher pressure in communication with the downhole fluid at the lower pressure through the piston to pressurize the

downhole fluid to a second higher pressure.

(6) Another embodiment may include a device or system for exchanging at least one property between fluids. The device may include at least one tank, at least one piston in the at least one tank, and a valve device. The tank may include a first end for receiving a first fluid (e.g., a clean fluid) with a first property and a second end for receiving a second fluid (e.g., a dirty fluid) with a second property. The piston may be configured to separate the clean fluid from the dirty fluid and to substantially prohibit fluid the dirty fluid from traveling from the second side to the first side.

(7) Another embodiment may include a piston for at least partially separating at least two fluid streams. The piston may be implemented in the devices or systems discussed above. The piston includes a body having an opening extending along an axis of the body where the opening defines a fluid pathway through the piston, and at least one valve obstructing the opening. The at least one valve is configured to enable fluid flow in one direction along the fluid pathway through the opening and to at least partially inhibit fluid flow in another, opposing direction along the fluid pathway through the opening.

(8) Another embodiment may include a method of operating a pressure exchange device comprising supplying a high pressure fluid to a high pressure inlet of a valve configured to direct flow of the high pressure fluid to a chamber; transferring a pressure from the high pressure fluid to a dirty fluid through a piston in the chamber; enabling some of the high pressure fluid to pass through the piston; and substantially prohibiting the dirty fluid from passing through the piston to the high pressure fluid.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) While the specification concludes with claims particularly pointing out and distinctly claiming what are regarded as embodiments of the present disclosure, various features and advantages of embodiments of the disclosure may be more readily ascertained from the following description of example embodiments of the disclosure when read in conjunction with the accompanying drawings, in which:

(2) FIG. 1 is schematic view of a hydraulic fracturing system according to an embodiment of the present disclosure;

(3) FIG. 2 is cross-sectional view of a fluid exchanger device according to an embodiment of the present disclosure;

(4) FIG. 3A is a cross-sectional view of a control valve in a first position according to an embodiment of the present disclosure;

(5) FIG. 3B is a cross-sectional view of a control valve in a second position according to an embodiment of the present disclosure;

(6) FIG. 4 is a isometric view of a piston according to an embodiment of the present disclosure; and

(7) FIG. 5 is a cross-sectional view of a piston according to an embodiment of the present disclosure.

DETAILED DESCRIPTION

(8) The illustrations presented herein are not meant to be actual views of any particular fluid exchanger or component thereof, but are merely idealized representations employed to describe illustrative embodiments. The drawings are not necessarily to scale. Elements common between figures may retain the same numerical designation.

(9) As used herein, relational terms, such as “first,” “second,” “top,” “bottom,” etc., are generally used for clarity and convenience in understanding the disclosure and accompanying drawings and do not connote or depend on any specific preference, orientation, or order, except where the context clearly indicates otherwise.

(10) As used herein, the term “and/or” means and includes any and all combinations of one or more of the associated listed items.

(11) As used herein, the terms “vertical” and “lateral” refer to the orientations as depicted in the figures.

(12) As used herein, the term “substantially” or “about” in reference to a given parameter means and includes to a degree that one skilled in the art would understand that the given parameter, property, or condition is met with a small degree of variance, such as within acceptable manufacturing tolerances. For example, a parameter that is substantially met may be at least 90% met, at least 95% met, at least 99% met, or even 100% met.

(13) As used herein, the term “fluid” may mean and include fluids of any type and composition. Fluids may take a liquid form, a gaseous form, or combinations thereof, and, in some instances, may include some solid material. In some embodiments, fluids may convert between a liquid form and a gaseous form during a cooling or heating process as described herein. In some embodiments, the term fluid includes gases, liquids, and/or pumpable mixtures of liquids and solids.

(14) Embodiments of the present disclosure may relate to exchange devices that may be utilized to exchange one or more properties between fluids (e.g., a pressure exchanger). Such exchangers (e.g., pressure exchangers) are sometimes called “flow-work exchangers” or “isobaric devices” and are machines for exchanging pressure energy from a relatively high-pressure flowing fluid system to a relatively low-pressure flowing fluid system.

(15) In some industrial processes, elevated pressures are required in certain parts of the operation to achieve the desired results, following which the pressurized fluid is depressurized. In other processes, some fluids used in the process are available at high-pressures and others at low-pressures, and it is desirable to exchange pressure energy between these two fluids. As a result, in some applications, great improvement in economy can be realized if pressure can be efficiently transferred between two fluids.

(16) In some embodiments, exchangers as disclosed herein may be similar to and include the various components and configurations of the pressure exchangers disclosed in U.S. Pat. No. 5,797,429 to Shumway, issued Aug. 25, 1998, the disclosure of which is hereby incorporated herein in its entirety by this reference.

(17) Although some embodiments of the present disclosure are depicted as being used and employed as a pressure exchanger between two or more fluids, persons of ordinary skill in the art will understand that the embodiments of the present disclosure may be employed in other implementations such as, for example, the exchange of other properties (e.g., temperature, density, etc.) and/or composition between one or more fluids and/or mixing of two or more fluids.

(18) In some embodiments, a pressure exchanger may be used to protect moving components (e.g., pumps, valves, impellers, etc.) in processes where high pressures are needed in a fluid that has the potential to damage the moving components (e.g., abrasive fluid, caustic fluid, acidic fluid, etc.).

(19) For example, pressure exchange devices according to embodiments of the disclosure may be implemented in hydrocarbon related processes, such as, hydraulic fracturing or other drilling operations (e.g., subterranean downhole drilling operations).

(20) As discussed above, well completion operations in the oil and gas industry often involve hydraulic fracturing, drilling operations, or other downhole operations that use high-pressure pumps to increase the pressure of the downhole fluid (e.g., fluid that is intended to be conducted into a subterranean formation or borehole, such as, fracking fluid, drilling fluid, drilling mud). The proppants, chemicals, additives to produce mud, etc. in these fluids often increase wear and maintenance on the high-pressure pumps.

(21) In some embodiments, a hydraulic fracturing system may include a hydraulic energy transfer system that transfers pressure between a first fluid (e.g., a clean fluid, such as a partially (e.g., majority) or substantially proppant free fluid or a pressure exchange fluid) and a second fluid (e.g., fracking fluid, such as a proppant-laden fluid, an abrasive fluid, or a dirty fluid). Such systems may

at least partially (e.g., substantially, primarily, entirely) isolate the high-pressure first fluid from the second dirty fluid while still enabling the pressurizing of the second dirty fluid with the high-pressure first fluid and without having to pass the second dirty fluid directly through a pump or other pressurizing device.

(22) While some embodiments discussed herein may be directed to fracking operations, in additional embodiments, the exchanger systems and devices disclosed herein may be utilized in other operations. For example, devices, systems, and/or method disclosed herein may be used in other downhole operations, such as, for example, downhole drilling operations. Further, embodiments of the pistons disclosed herein may be implemented in any other suitable fluid-handling application.

(23) FIG. 1 illustrates a system diagram of an embodiment of hydraulic fracturing system **100** utilizing a pressure exchanger between a first fluid stream (e.g., clean fluid stream) and a second fluid stream (e.g., a fracking fluid stream). Although not explicitly described, it should be understood that each component of the system **100** may be directly connected or coupled via a fluid conduit (e.g., pipe) to an adjacent (e.g., upstream or downstream) component. The hydraulic fracturing system **100** may include one or more devices for pressurizing the first fluid stream, such as, for example, frack pumps **102** (e.g., reciprocating pumps, centrifugal pumps, scroll pumps, etc.). The system **100** may include multiple frack pumps **102**, such as at least two frack pumps **102**, at least four frack pumps **102**, at least ten frack pumps **102**, at least sixteen frack pumps, or at least twenty frack pumps **102**. In some embodiments, the frack pumps **102** may provide relatively and substantially clean fluid at a high pressure to a pressure exchanger **104** from a fluid source **101**. In some embodiments, fluid may be provided separately to each pump **102** (e.g., in a parallel configuration). After pressurization in the pumps **102**, the high pressure clean fluid **110** may be combined and transmitted to the pressure exchanger **104** (e.g., in a serial configuration).

(24) As used herein, “clean” fluid may describe fluid that is at least partially or substantially free (e.g., substantially entirely or entirely free) of chemicals and/or proppants typically found in a downhole fluid and “dirty” fluid may describe fluid that at least partially contains chemicals, other additives and/or proppants typically found in a downhole fluid.

(25) The pressure exchanger **104** may transmit the pressure from the high pressure clean fluid **110** to a low pressure fracking fluid (e.g., fracking fluid **112**) in order to provide a high pressure fracking fluid **116**. The clean fluid may be expelled from the pressure exchanger **104** as a low pressure fluid **114** after the pressure is transmitted to the low pressure fracking fluid **112**. In some embodiments, the low pressure fluid **114** may be an at least partially or substantially clean fluid that substantially lacks chemicals and/or proppants aside from a small amount that may be passed to the low pressure fluid **114** from the fracking fluid **112** in the pressure exchanger **104**.

(26) In some embodiments, the pressure exchanger **104** may include one or more pressure exchanger devices (e.g., operating in parallel). In such configurations, the high pressure inputs may be separated and provided to inputs of each of the pressure exchanger devices. The outputs of each of the pressure exchanger devices may be combined as the high pressure fracking fluid exits the pressure exchanger **104**. For example, and as discussed below with reference to FIG. 4, the pressure exchanger **104** may include two or more (e.g., three) pressure exchanger devices operating in parallel. As depicted, the pressure exchanger **104** may be provided on a mobile platform (e.g., a truck trailer) that may be relatively easily installed and removed from a fracking well site.

(27) After being expelled from the pressure exchanger **104**, the low pressure clean fluid **114** may travel to and be collected in a mixing chamber **106** (e.g., blender unit, mixing unit, etc.). In some embodiments, the low pressure fluid **114** may be converted (e.g., modified, transformed, etc.) to the low pressure fracking fluid **112** in the mixing chamber **106**. For example, a proppant may be added to the low pressure clean fluid **114** in the mixing chamber **106** creating a low pressure fracking fluid **112**. In some embodiments, the low pressure clean fluid **114** may be expelled as waste.

(28) In many hydraulic fracturing operations, a separate process may be used to heat the fracking

fluid **112** before the fracking fluid **112** is discharged downhole (e.g., to ensure proper blending of the proppants in the fracking fluid). In some embodiments, using the low pressure clean fluid **114** to produce the fracking fluid **112** may eliminate the step of heating the fracking fluid. For example, the low pressure clean fluid **114** may be at an already elevated temperature as a result of the fracking pumps **102** pressurizing the high pressure clean fluid **110**. After transferring the pressure in the high pressure clean fluid **110** that has been heated by the pumps **102**, the now low pressure clean fluid **114** retains at least some of that heat energy as it is passed out of the pressure exchanger **104** to the mixing chamber **106**. In some embodiments, using the low pressure clean fluid **114** at an already elevated temperature to produce the fracking fluid may result in the elimination of the heating step for the fracking fluid. In other embodiments, the elevated temperature of the low pressure clean fluid **114** may result in a reduction of the amount of heating required for the fracking fluid.

(29) After the proppant is added to the low pressure fluid **114**, now fracking fluid, the low pressure fracking fluid **112** may be expelled from the mixing chamber **106**. The low pressure fracking fluid **112** may then enter the pressure exchanger **104** on the fracking fluid end through a fluid conduit **108** connected (e.g., coupled) between the mixing chamber **106** and the pressure exchanger **104**. Once in the pressure exchanger **104**, the low pressure fracking fluid **112** may be pressurized by the transmission of pressure from the high pressure clean fluid **110** through the pressure exchanger **104**. The high pressure fracking fluid **116** may then exit the pressure exchanger **104** and be transmitted downhole.

(30) Hydraulic fracturing systems generally require high operating pressures for the high pressure fracking fluid **116**. In some embodiments, the desired pressure for the high pressure fracking fluid **116** may be between about 8,000 PSI (55,158 kPa) and about 12,000 PSI (82,737 kPa), such as between about 9,000 PSI (62,052 kPa) and about 11,000 PSI (75,842 kPa), or about 10,000 PSI (68,947 kPa).

(31) In some embodiments, the high pressure clean fluid **110** may be pressurized to a pressure at least substantially the same or slightly greater than the desired pressure for the high pressure fracking fluid **116**. For example, the high pressure clean fluid **110** may be pressurized to between about 0 PSI (0 kPa) and about 1000 PSI (6,894 kPa) greater than the desired pressure for the high pressure fracking fluid **116**, such as between about 200 PSI (1,379 kPa) and about 700 PSI (4,826 kPa) greater than the desired pressure, or between about 400 PSI (2,758 kPa) and about 600 PSI (4,137 kPa) greater than the desired pressure, to account for any pressure loss during the pressure and exchange process.

(32) FIG. 2 illustrates an embodiment of a pressure exchanger **200**. The pressure exchanger **200** may be a linear pressure exchanger in the sense that it is operated by moving or translating an actuation assembly substantially along a linear path. For example, the actuation assembly may be moved linearly to selectively place the low and high pressure fluids in at least partial communication (e.g., indirect communication where the pressure of the high pressure fluid may be transferred to the low pressure fluid) as discussed below in greater detail.

(33) The linear pressure exchanger **200** may include one or more (e.g., two) chambers **202a**, **202b** (e.g., tanks, collectors, cylinders, tubes, pipes, etc.). The chambers **202a**, **202b** (e.g., parallel chambers **202a**, **202b**) may include pistons **204a**, **204b** configured to substantially maintain the high pressure clean fluid **210** and low pressure clean fluid **214** (e.g., the clean side) separate from the high pressure dirty fluid **216** and the low pressure dirty fluid **212** (e.g., the dirty side) while enabling transfer of pressure between the respective fluids **210**, **212**, **214**, and **216**. The pistons **204a**, **204b** may be sized (e.g., the outer diameter of the pistons **204a**, **204b** relative to the inner diameter of the chambers **202a**, **202b**) to enable the pistons **204a**, **204b** to travel through the chamber **202a**, **202b** while minimizing fluid flow around the pistons **204a**, **204b**.

(34) The linear pressure exchanger **200** may include a clean control valve **206** configured to control the flow of high pressure clean fluid **210** and low pressure clean fluid **214**. Each of the chambers

202a, 202b may include one or more dirty control valves **207a, 207b, 208a, and 208b** configured to control the flow of the low pressure dirty fluid **212** and the high pressure dirty fluid **216**.

(35) While the embodiment of FIG. 2 contemplates a linear pressure exchanger **200**, other embodiments, may include other types of pressure exchangers that involve other mechanisms for selectively placing the low and high pressure fluids in at least partial communication (e.g., a rotary actuator such as those disclosed in U.S. Pat. No. 9,435,354, issued Sep. 6, 2016, the disclosure of which is hereby incorporated herein in its entirety by this reference, etc.).

(36) In some embodiments, the clean control valve **206**, which includes an actuation stem **203** that moves one or more stoppers **308** along (e.g., linearly along) a body **205** of the valve **206**, may selectively allow (e.g., input, place, etc.) high pressure clean fluid **210** provided from a high pressure inlet port **302** to enter a first chamber **202a** on a clean side **220a** of the piston **204a**. The high pressure clean fluid **210** may act on the piston **204a** moving the piston **204a** in a direction toward the dirty side **221a** of the piston **204a** and compressing the dirty fluid in the first chamber **202a** to produce the high pressure dirty fluid **216**. The high pressure dirty fluid **216** may exit the first chamber **202a** through the dirty discharge control valve **208a** (e.g., outlet valve, high pressure outlet). At substantially the same time, the low pressure dirty fluid **212** may be entering the second chamber **202b** through the dirty fill control valve **207b** (e.g., inlet valve, low pressure inlet). The low pressure dirty fluid **212** may act on the dirty side **221b** of the piston **204b** moving the piston **204b** in a direction toward the clean side **220b** of the piston **204b** in the second chamber **202b**. The low pressure clean fluid **214** may be discharged (e.g., emptied, expelled, etc.) through the clean control valve **206** as the piston **204b** moves in a direction toward the clean side **220b** of the piston **204b** reducing the space on the clean side **220b** of the piston **204b** within the second chamber **202b**. A cycle of the pressure exchanger is completed once each piston **204a, 204b** moves the substantial length (e.g., the majority of the length) of the respective chamber **202a, 202b** (which “cycle” may be a half cycle with the piston **204a, 204b** moving in one direction along the length of the chamber **202a, 202b** and a full cycle includes the piston **204a, 204b** moving in the one direction along the length of the chamber **202a, 202b** and then moving in the other direction to return to substantially the original position). In some embodiments, only a portion of the length may be utilized (e.g., in reduced capacity situations). Upon the completion of a cycle, the actuation stem **203** of the clean control valve **206** may change positions enabling the high pressure clean fluid **210** to enter the second chamber **202b**, thereby changing the second chamber **202b** to a high pressure chamber and changing the first chamber **202a** to a low pressure chamber and repeating the process.

(37) In some embodiments, each chamber **202a, 202b** may have a higher pressure on one side of the pistons **204a, 204b** to move the piston in a direction away from the higher pressure. For example, the high pressure chamber may experience pressures between about 8,000 PSI (55,158 kPa) and about 13,000 PSI (89,632 kPa) with the highest pressures being in the high pressure clean fluid **210** to move the piston **204a, 204b** away from the high pressure clean fluid **210** compressing and discharging the dirty fluid to produce the high pressure dirty fluid **216**. The low pressure chamber **202a, 202b** may experience much lower pressures, relatively, with the relatively higher pressures in the currently low pressure chamber **202a, 202b** still being adequate enough in the low pressure dirty fluid **212** to move the piston **204a, 204b** in a direction away from the low pressure dirty fluid **212** discharging the low pressure clean fluid **214**. In some embodiments, the pressure of the low pressure dirty fluid **212** may be between about 100 PSI (689 kPa) and about 700 PSI (4,826 kPa), such as between about 200 PSI (1,379 kPa) and about 500 PSI (3,447 kPa), or between about 300 PSI (2,068 kPa) and about 400 PSI (2,758 kPa).

(38) Referring back to FIG. 1, in some embodiments, the system **100** may include an optional device (e.g., a pump) to pressurize the low pressure dirty fluid **212** (e.g., to a pressure level that is suitable to move the piston **204a, 204b** toward the clean side) as it is being provided into the chambers **202a, 202b**.

(39) Referring again to FIG. 2, if any fluid pushes past the piston **204a, 204b** (e.g., leak by, blow

by, etc.) it will generally tend to flow from the higher pressure fluid to the lower pressure fluid. The high pressure clean fluid **210** may be maintained at the highest pressure in the system such that the high pressure clean fluid **210** may not generally become substantially contaminated. The low pressure clean fluid **214** may be maintained at the lowest pressure in the system. Therefore, it is possible that the low pressure clean fluid **214** may become contaminated by the low pressure dirty fluid **212**. In some embodiments, the low pressure clean fluid **214** may be used to produce the low pressure dirty fluid **212** substantially nullifying any detriment resulting from the contamination. Likewise, any contamination of the high pressure dirty fluid **216** by the high pressure clean fluid **210** would have minimal effect on the high pressure dirty fluid **216**.

(40) In some embodiments, the dirty control valves **207a**, **207b**, **208a**, **208b** may be check valves (e.g., clack valves, non-return valves, reflux valves, retention valves, or one-way valves). For example, one or more of the dirty control valves **207a**, **207b**, **208a**, **208b** may be a ball check valve, diaphragm check valve, swing check valve, tilting disc check valve, clapper valve, stop-check valve, lift-check valve, in-line check valve, duckbill valve, etc. In additional embodiments, one or more of the dirty control valves **207a**, **207b**, **208a**, **208b** may be actuated valves (e.g., solenoid valves, pneumatic valves, hydraulic valves, electronic valves, etc.) configured to receive a signal from a controller and open or close responsive the signal.

(41) The dirty control valves **207a**, **207b**, **208a**, **208b** may be arranged in opposing configurations such that when the chamber **202a**, **202b** is in the high pressure configuration the high pressure dirty fluid opens the dirty discharge control valve **208a**, **208b** while the pressure in the chamber **202a**, **202b** holds the dirty fill control valve **207a**, **207b** closed. For example, the dirty discharge control valve **208a**, **208b** comprises a check valve that opens in a first direction out of the chamber **202a**, **202b**, while the dirty fill control valve **207a**, **207b** comprises a check valve that opens in a second, opposing direction into the chamber **202a**, **202b**.

(42) The dirty discharge control valves **208a**, **208b** may be connected to a downstream element (e.g., a fluid conduit, a separate or common manifold) such that the high pressure in the downstream element holds the dirty discharge valve **208a**, **208b** closed in the chamber **202a**, **202b** that is in the low pressure configuration. Such a configuration enables the low pressure dirty fluid to open the dirty fill control valve **207a**, **207b** and enter the chamber **202a**, **202b**.

(43) FIGS. 3A and 3B illustrate a cross sectional view of an embodiment of a clean control valve **300** at two different positions. In some embodiments, the clean control valve **300** may be similar to the control valve **206** discussed above. The clean control valve **300** may be a multiport valve (e.g., 4 way valve, 5 way valve, LinX® valve, etc.). The clean control valve **300** may have one or more high pressure inlet ports (e.g., one port **302**), one or more low pressure outlet ports (e.g., two ports **304a**, **304b**), and one or more chamber connection ports (e.g., two ports **306a**, **306b**). The clean control valve **300** may include at least two stoppers **308** (e.g., plugs, pistons, discs, valve members, etc.). In some embodiments, the clean control valve **300** may be a linearly actuated valve. For example, the stoppers **308** may be linearly actuated such that the stoppers **308** move along a substantially straight line (e.g., along a longitudinal axis L.sub.300 of the clean control valve **300**).

(44) The clean control valve **300** may include an actuator **303** configured to actuate the clean control valve **300** (e.g., an actuator coupled to a valve stem **301** of the clean control valve **300**). In some embodiments, the actuator **303** may be electronic (e.g., solenoid, rack and pinion, ball screw, segmented spindle, moving coil, etc.), pneumatic (e.g., tie rod cylinders, diaphragm actuators, etc.), or hydraulic. In some embodiments, the actuator **303** may enable the clean control valve **300** to move the valve stem **301** and stoppers **308** at variable rates (e.g., changing speeds, adjustable speeds, etc.).

(45) FIG. 3A illustrates the clean control valve **300** in a first position. In the first position, the stoppers **308** may be positioned such that the high pressure clean fluid may enter the clean control valve **300** through the high pressure inlet port **302** and exit into a first chamber through the chamber connection port **306a**. In the first position, the low pressure clean fluid may travel through

the clean control valve **300** between the chamber connection port **306b** and the low pressure outlet port **304b** (e.g., may exit through the low pressure outlet port **304b**).

(46) FIG. **3B** illustrates the clean control valve **300** in a second position. In the second position, the stoppers **308** may be positioned such that the high pressure clean fluid may enter the clean control valve **300** through the high pressure inlet port **302** and exit into a second chamber through the chamber connection port **306b**. The low pressure clean fluid may travel through the clean control valve **300** between the chamber connection port **306a** and the low pressure outlet port **304a** (e.g., may exit through the low pressure outlet port **304a**).

(47) Now referring to FIGS. **2**, **3A**, and **3B**, the clean control valve **206** is illustrated in the first position with the high pressure inlet port **302** connected to the chamber connection port **306a** providing high pressure clean fluid to the first chamber **202a**. Upon completion of the cycle, the clean control valve **206** may move the stoppers **308** to the second position thereby connecting the high pressure inlet port **302** to the second chamber **202b** through the chamber connection port **306b**.

(48) In some embodiments, the clean control valve **206** may pass through a substantially fully closed position in the middle portion of a stroke between the first position and the second position. For example, in the first position, the stoppers **308** may maintain a fluid pathway between the high pressure inlet port **302** and the chamber connection port **306a** and a fluid pathway between the chamber connection port **306b** and the low pressure outlet port **304b**. In the second position, the stoppers **308** may maintain a fluid pathway between the high pressure inlet port **302** and the chamber connection port **306b** and a fluid pathway between the chamber connection port **306a** and the low pressure outlet port **304a**. Transitioning between the first and second positions may involve at least substantially closing both fluid pathways to change the connection of the chamber connection port **306a** from the high pressure inlet port **302** to the low pressure outlet port **304a** and to change the connection of the chamber connection port **306b** from the low pressure outlet port **304b** to the high pressure inlet port **302**. The fluid pathways may at least substantially close at a middle portion of the stroke to enable the change of connections. Opening and closing valves, where fluids are operating at high pressures may result in pressure pulsations (e.g., water hammer) that can result in damage to components in the system when high pressure is suddenly introduced or removed from the system. As a result, pressure pulsations may occur in the middle portion of the stroke when the fluid pathways are closing and opening respectively.

(49) In some embodiments, the actuator **303** may be configured to move the stoppers **308** at variable speeds along the stroke of the clean control valve **206**. As the stoppers **308** move from the first position to the second position, the stoppers **308** may move at a high rate of speed while traversing a first portion of the stroke that does not involve newly introducing flow from the high pressure inlet port **302** into the chamber connection ports **306a**, **306b**. The stoppers **308** may decelerate to a low rate of speed as the stoppers **308** approach a closed position (e.g., when the stoppers **308** block the chamber connection ports **306a**, **306b** during the transition between the high pressure inlet port **302** connection and the low pressure outlet port **304a**, **304b** connection) at a middle portion of the stroke. The stoppers **308** may continue at a lower rate of speed, as the high pressure inlet port **302** is placed into communication with one of the chamber connection ports **306a**, **306b**. After, traversing the chamber connection ports **306a**, **306b**, the stoppers **308** may accelerate to another high rate of speed as the stoppers **308** approach the second position. The low rate of speed in the middle portion of the stroke may reduce the speed that the clean control valve **206** opens and closes enabling the clean control valve to gradually introduce and/or remove the high pressure from the chambers **202a**, **202b**.

(50) In some embodiments, the motion of the pistons **204a**, **204b** may be controlled by regulating the rate of fluid flow (e.g., of the incoming fluid) and/or a pressure differential between the clean side **220a**, **220b** of the pistons **204a**, **204b**, and the dirty side **221a**, **221b** of the pistons **204a**, **204b** at least partially with the movement of the clean control valve **206**. In some embodiments, it may

be desirable for the piston **204a**, **204b** in the low pressure chamber **202a**, **202b** to move at substantially the same speed as the piston **204a**, **204b** in the high pressure chamber **202a**, **202b** either by manipulating their pressure differentials in each chamber and/or by controlling the flow rates of the fluid in and out of the chambers **202a**, **202b**. However, the piston **204a**, **204b** in the low pressure chamber **202a**, **202b** may tend to move at a greater speed than the piston **204a**, **204b** in the high pressure chamber **202a**, **202b**.

(51) In some embodiments, the rate of fluid flow and/or the pressure differential may be varied to control acceleration and deceleration of the pistons **204a**, **204b** (e.g., by manipulating and/or varying the stroke of the clean control valve **206** and/or by manipulating the pressure in the fluid streams with one or more pumps). For example, increasing the flow rate and/or the pressure of the high pressure clean fluid **210** when the piston **204a**, **204b** is near a clean end **224** of the chamber **202a**, **202b** at the beginning of the high pressure stroke may increase the rate of fluid flow and/or the pressure differential in the chamber **202a**, **202b**. Increasing the rate of fluid flow and/or the pressure differential may cause the piston **204a**, **204b** to accelerate to or move at a faster rate. In another example, the flow rate and/or the pressure of the high pressure clean fluid **210** may be decreased when the piston **204a**, **204b** approaches a dirty end **226** of the chamber **202a**, **202b** at the end of the high pressure stroke. Decreasing the rate of fluid flow and/or the pressure differential may cause the piston **204a**, **204b** to decelerate and/or stop before reaching the dirty end of the respective chamber **202a**, **202b**.

(52) Similar control with the stroke of the clean control valve **206** may be utilized to prevent the piston **204a**, **204b** from traveling to the furthest extent of the clean end of the chambers **202a**, **202b**. For example, the clean control valve **206** may close off one of the chamber connection ports **306a**, **306b** before the piston **204a**, **204b** contacts the furthest extent of the clean end of the chambers **202a**, **202b** by preventing any further fluid flow and slowing and/or stopping the piston **204a**, **204b**. In some embodiments, the clean control valve **206** may open one of the chamber connection ports **306a**, **306b** into communication with the high pressure inlet port **302** before the piston **204a**, **204b** contacts the furthest extent of the clean end of the chambers **202a**, **202b** in order to slow, stop, and/or reverse the motion of the piston **204a**, **204b**.

(53) If the pistons **204a**, **204b** reach the clean end **224** or dirty end **226** of the respective chambers **202a**, **202b** the higher pressure fluid may bypass the piston **204a**, **204b** and mix with the lower pressure fluid. In some embodiments, mixing the fluids may be desirable. For example, if the pistons **204a**, **204b** reach the dirty end **226** of the respective chambers **202a**, **202b** during the high pressure stroke, the high pressure clean fluid **210** may bypass the piston **204a**, **204b** (e.g., by traveling around the piston **204a**, **204b** or through a valve in the piston **204a**, **204b**) flushing any residual contaminants from the surfaces of the piston **204a**, **204b**. In some embodiments, mixing the fluids may be undesirable. For example, if the pistons **204a**, **204b** reach the clean end **224** of the respective chambers **202a**, **202b** during the low pressure stroke, the low pressure dirty fluid **212** may bypass the piston **204a**, **204b** and mix with the low pressure clean fluid contaminating the clean area in the clean control valve **206** with the dirty fluid.

(54) In some embodiments, the system **100** may prevent the pistons **204a**, **204b** from reaching the clean end **224** of the respective chambers **202a**, **202b**. For example, the clean control valve **206** may include a control device **209** (e.g., sensor, safety, switch, etc.) to trigger the change in position of the clean control valve **206** on detecting the approach of the piston **204a**, **204b** to the clean end **224** of the respective chamber **202a**, **202b** such that the system **100** may utilize the clean control valve **206** to change flow path positions before the piston **204a**, **204b** reaches the clean end **224** of the chamber **202a**, **202b**.

(55) In some embodiments, the system **100** may be configured to enable the pistons **204a**, **204b** to reach the dirty end **226** of the respective chambers **202a**, **202b** during the high pressure stroke. In some embodiments, the clean control valve **206** may include a control device **209** to trigger the change in position of the clean control valve **206** on detecting the approach of the piston **204a**,

204b to the dirty end **226** of the respective chamber **202a, 202b**. In some embodiments, the control device may be configured such that the control valve **206** does not complete the change in direction of the piston **204a, 204b** until the piston **204a, 204b** has reached the furthest extent of the dirty end **226** of the respective chamber **202a, 202b**. In some embodiments, the control device may include a time delay through programming or mechanical delay that enables the piston **204a, 204b** to reach the furthest extent of the dirty end **226** of the chamber **202a, 202b**.

(56) In some embodiments, the system **100** may be configured to enable the pistons **204a, 204b** to reach the dirty end **226** of the respective chambers **202a, 202b** during the high pressure stroke and prevent the pistons **204a, 204b** from reaching the clean end **224** of the respective chambers **202a, 202b** during the low pressure stroke. For example, the system **100** may drive both of the pistons **204a, 204b** a select distance through the respective chambers **202a, 202b** where the pistons **204a, 204b** is maintained a select distance from the clean end **224** while enabling the pistons **204a, 204b** to travel relatively closer to or come in contact with, the dirty end **226**. In some embodiments, the system **100** may be configured such that the pressure differential across the piston **204a, 204b** in the low pressure chamber **202a, 202b** may be less than the pressure differential across the piston **204a, 204b** in the high pressure chamber **202a, 202b** such that the piston **204a, 204b** travels slower during the low pressure cycle than the high pressure cycle.

(57) In some embodiments, the control device **209** may be configured to trigger the change in position of the clean control valve **206** on detecting the approach of the piston **204a, 204b** to the clean end **224** of the respective chamber **202a, 202b** such that the clean control valve **206** may change positions before the piston **204a, 204b** reaches the clean end **224** of the chamber **202a, 202b**. In some embodiments, the control device **209** may be configured to trigger the change in position of the clean control valve **206** on detecting the approach of the piston **204a, 204b** to the dirty end **226** of the respective chamber **202a, 202b**. In some embodiments, the control device may be configured to trigger the change in position of the clean control valve **206** by evaluating both of the pistons **204a, 204b** as they respectively approach the clean end **224** and the dirty end **226** of the chambers **202a, 202b**. For example, the control device **209** may detect the approach of the piston **204a, 204b** to the dirty end **226** of the chamber **202a, 202b** and begin a timer (e.g., mechanical timer, electronic timer, programmed time delay, etc.) If the control device **209** detects the approach of the piston **204a, 204b** to the clean end **224** of the chamber **202a, 202b** before the time triggers the change in position of the clean control valve **206**, the control device **209** may override the timer and change the position of the clean control valve **206** to prevent the piston **204a, 204b** from reaching the clean end **224** of the chamber **202a, 202b**.

(58) In some embodiments, an automated controller may produce signals that may be transmitted to the clean control valve **206** directing the clean control valve **206** to move from the first position to the second position or from the second position to the first position (e.g., at a constant and/or variable rate).

(59) Referring back to FIG. 2, the pressure exchanger **200** may include one or more bypass features **310** (e.g., in one or more of the chambers **202a, 202b** and/or in one or more of the pistons **204a, 204b**) for bypassing fluid around and/or through the pistons **204a, 204b**. For example, the pistons **204a, 204b** may include a bypass feature **310** for enabling fluid to pass through the pistons **204a, 204b**. In some embodiments, the bypass feature **310** may be associated with each of the pistons **204a, 204b** (e.g., positioned in the pistons **204a, 204b**) where the bypass feature **310** comprises a valve, such as a one-way check valve, or another type of suitable valve, such as those listed above, that may limit flow in one or more directions while enabling flow in one or more other directions.

(60) In additional embodiments, the chambers **202a, 202b** may define all or a portion of the bypass feature **310**. For example, an entirety of the bypass feature **310** may be in the chambers **202a, 202b** or may be defined partially by the chambers **202a, 202b** and partially by the pistons **204a, 204b**.

(61) As discussed above, the system **100** may enable the pistons **204a, 204b** to reach an end (e.g., the dirty end **226**) of the respective chambers **202a, 202b** during a stroke. In such an event, the

bypass feature **310** may enable fluid to the pass through the bypass feature **310** in order to reach one or more of the control valves **207a**, **207b**, **208a**, and **208b** (e.g., in an underflush in supplying an overflow of high pressure fluid into the chambers **202a**, **202b**). For example, the system **100** may drive both of the pistons **204a**, **204b** a select distance through the respective chambers **202a**, **202b** enabling the pistons **204a**, **204b** to travel closer to, or come in contact with, the end **226**. As the pistons **204a**, **204b** approach and/or come in contact with the end **226**, the bypass feature **310** may be activated (e.g., by the overcoming of a biasing feature) and begin to enable fluid to pass around and/or through the bypass feature **310**. For example, the obstruction of the end **226** of the chambers **202a**, **202b** and/or increasing fluid pressure as the pistons **204a**, **204b** approach the end **226** may resist further movement of the pistons **204a**, **204b**, applying an opposing force to the pistons **204a**, **204b**. As the force of the fluid flow is no longer being substantially utilized to move the pistons **204a**, **204b**, the fluid may then overcome the bypass feature **310** (e.g., when a selected differential pressure is exceeded) in order to enable fluid flow past the pistons **204a**, **204b**.

(62) Is such an embodiment, the bypass feature **310** may be utilized to over supply fluid (e.g., clean fluid) on one side of the pistons **204a**, **204b** into the chambers **202a**, **202b**. The fluid may pass through the bypass feature **310** of the pistons **204a**, **204b** to another side of the pistons **204a**, **204b**.

(63) As also discussed above, the system **100** may prevent the pistons **204a**, **204b** from reaching an end (e.g., the clean end **224**) of the respective chambers **202a**, **202b** during a stroke. However, even if the pistons **204a**, **204b** approach or reach the end **224**, the bypass feature **310** may prevent fluid from passing through the bypass feature **310** in order to, for example, reach one or more of the clean control valve **206** (e.g., in an oversupply of low pressure dirty fluid into the chambers **202a**, **202b**).

(64) The pistons **204a**, **204b** (e.g., an outer diameter of the pistons **204a**, **204b**) may be sized relative to an inner dimension of the chambers **202a**, **202b** to enable the pistons **204a**, **204b** to pass through the chambers **202a**, **202b** without binding while minimizing leakage around the pistons **204a**, **204b** (e.g., minimizing contamination of the clean fluid). The geometry of pistons **204a**, **204b** relative to an inner dimension of the chambers **202a**, **202b** may minimize proppants (e.g., sand) in the dirty fluid from traveling, and becoming trapped, between the pistons **204a**, **204b** to pass through the chambers **202a**, **202b**.

(65) In some embodiments, the pistons **204a**, **204b** may include materials that are abrasion resistant or that include an abrasion resistant coating (e.g., ceramics, carbides, metals, polymers, etc.). In addition, the material of the pistons **204a**, **204b** may be selected to exhibit a medium density. Such an embodiment may provide at least some buoyancy of the pistons **204a**, **204b** in fluid, thereby reducing wear on the bottom of the chambers **202a**, **202b** that may occur due to the gravity of the pistons **204a**, **204b**.

(66) FIG. 4 is an isometric view of a piston **400** with a portion of a bypass feature **401** shown in a transparent view for clarity. In some embodiments, the piston **400** and the bypass feature **401** may be similar to, and include one or more components of, the pistons **204a**, **204b** and the bypass features **310** discussed above in relation to FIG. 2. As shown in FIG. 4, the piston **400** includes an outer sidewall **402** (e.g., a radial sidewall) defining an outermost extent (e.g., an outer diameter) the piston **400**.

(67) The outer sidewall **402** may surround (e.g., encircle) an inner portion of the piston **400** including the bypass feature **401**. As depicted, the piston **400** may include a first recess **404** defined in the piston **400** such that the bypass feature **401** is positioned at a central portion **406** of the piston **400** at a base of the recess **404**. The central portion **406** of the piston **400** may be defined as a web with openings **408** (e.g., one, two three, four, or more openings or channels through the central portion **406** of the piston **400**) that enable fluid to pass through the piston **400** when the bypass feature **401** is an open position enabling fluid flow through the piston **400**.

(68) FIG. 5 is a cross-sectional view of a piston (e.g., the piston **400** shown in FIG. 4). As shown in FIG. 5, the piston **400** includes a first axial end **410** and a second, opposing axial end **412**. The first

recess **404** (e.g., having a cylindrical, conical, or frustoconical shape) may extend from the first axial end **410** into the piston **400** to the central portion **406**. A second recess **414** (e.g., having a cylindrical, conical, or frustoconical shape) may extend from the second axial end **412** into the piston **400** to the central portion **406**.

(69) The openings **408** at the central portion **406** of the piston **400** may extend between and connect (e.g., place in fluid communication) the first and second recesses **404**, **414**.

(70) The bypass feature **401** may be positioned at the central portion **406** of the piston **400** between the first and second recesses **404**, **414**. As depicted, the bypass feature **401** may include a valve member **416** that contacts and defines a seal between a portion of the piston **400** (e.g., a seating portion **418** of the central portion **406**) in the first recess **404**. As depicted, the bypass feature **401** may comprise a one-way check valve. As noted above, in additional embodiments, the bypass feature **401** may comprise another fluid flow restriction feature, such as, a multiple-way check valve or other type of valve.

(71) In a closed position (e.g., as shown in FIG. 5), the valve member **416** may contact the seating portion **418** and at least partially prevent (e.g., substantially entirely prevent) fluid from passing through the openings **408** to the second recess **414**. For example, in the closed position, the valve member **416** may prevent fluid flow from the first axial end **410** to the second axial end **412**.

(72) In some embodiments, the valve member **416** may be biased into the closed position. For example, the valve member **416** may be coupled to a biasing feature (e.g., a spring **420**, such as, a compression spring, a torsion spring, a Belleville spring, a constant force spring, a extension spring, etc.) on a mounting structure **422** (e.g., a rod coupled to the valve member **416** and the spring **420**). The mounting structure **422** may enable the valve member **416** to move (e.g., along an axis of the piston **400**) to an open position while compressing the spring **420** under the force of fluid flow from the second axial end **412** to the first axial end **410**. When the force of the spring **420** overcomes the force of the fluid flow, the spring **420** may force the valve member **416** back into the closed position.

(73) In the open position, the valve member **416** may move (e.g., axially displace) from the seating portion **418** and enable fluid to pass through the openings **408** from the second recess **414** to the first recess **404**. For example, clean fluid may pass from the second axial end **412** to the first axial end **410**, which first axial end **410** may be positioned on a dirty side of the piston **400** in communication with a dirty fluid. In embodiments with a biasing feature, movement of the valve member **416** from the seating portion **418** may compress the spring **420**.

(74) In some embodiments, the dimension and/or geometry of the piston **400** (e.g., proximate the outermost diameter of the piston **400** may be selected to minimize abrasion of the piston **400** and the chamber **202a**, **202b** (FIG. 2). For example, the piston **400** may include a chamfered edge or surface **424** (e.g., a chamfered cylindrical wall) at one or both of the first axial end **410** and the second axial end **412**. The chamfered surface **424** may extend from a respective axial end **410**, **412** to the outer sidewall **402**. As depicted, an entirety of the outermost surface of the piston **400** at both axial ends **410**, **412** may comprise an arcuate continuous surface leading from the axial ends **410**, **412** to the outer sidewall **402**. In some embodiments, the piston **400** including the chamfered surfaces **424** may act to create fluid vortices at the chamfered surfaces **424** (e.g., an interface between the piston **400** and an inner wall of the chamber **202a**, **202b**) that may suspend particles in the fluid through which the piston **400** is traveling in order to at least partially prevent the particles from becoming wedged between the piston **400** and the chamber **202a**, **202b** (FIG. 2).

(75) As discussed above, the piston **400** may be utilized in a system (e.g., system **100**) that may sense or otherwise monitor the position of the piston **400**. In some embodiments, the piston **400** may include one or more position detection features that enable detection of the presence of the piston **400** through, for example, a sensor (e.g., a contact or contactless sensor, such as a magnetic sensor, an optical sensor, an inductive proximity sensors, a Hall Effect sensor, an ultrasonic sensor, a capacitive proximity sensors, a contact, a button, a switch, etc.). For example, the piston **400** may

include one or more position detection features comprising one or more magnets **426** (e.g., relatively strong permanent magnets) positioned about the piston **400** (e.g., spaced at intervals about the circumference of the piston **400**). The magnetic field produced the magnets **426** may be detected by a complementary sensor (e.g., part of the control system of linear pressure exchanger **200** (FIG. 2)) to determine a position of the piston **400** (e.g., detect the passing of the piston **400** by the sensor).

(76) Pressure exchangers may reduce the amount of wear experienced by high pressure pumps, turbines, and valves in systems with abrasive, caustic, or acidic fluids. The reduced wear may allow the systems to operate for longer periods with less down time resulting in increased revenue or productivity for the systems. Additionally, the repair costs may be reduced as fewer parts may wear out. In operations such as fracking operations, where abrasive fluids are used at high temperatures, repairs and downtime can result in millions of dollars of losses in a single operation. Embodiments of the present disclosure may result in a reduction in wear experienced by the components of systems where abrasive, caustic, or acidic fluids are used at high temperatures. The reduction in wear will result in cost reduction and increased revenue production.

(77) Pistons in accordance with some embodiments of the instant disclosure may act to minimize fluid from one side of the piston (e.g., a fracking fluid) from contaminating fluid (e.g., clean fluid) from the other side of the piston. Further, the piston may enable fluid flow through a valve of the piston to enable fluid in one direction (e.g., enabling clean fluid to flow to a dirty side of the piston) while prohibiting fluid flow in another direction (e.g., preventing dirty fluid from traveling to a clean side of the piston). The geometry and/or material of the piston may further reduce abrasion on the piston as it moves through a chamber while minimizing unwanted fluid flow around the piston and particle from becoming stuck around an outer circumference of the piston. Further, the geometry may create fluid vortices that may suspend particles in the fluid through which the piston is traveling in order to at least partially prevent the particles from becoming wedged between the piston and the chamber. Finally, the piston may include one or more features for enabling the detection of the position of the piston.

(78) While the present disclosure has been described herein with respect to certain illustrated embodiments, those of ordinary skill in the art will recognize and appreciate that it is not so limited. Rather, many additions, deletions, and modifications to the illustrated embodiments may be made without departing from the scope of the disclosure as hereinafter claimed, including legal equivalents thereof. In addition, features from one embodiment may be combined with features of another embodiment while still being encompassed within the scope of the disclosure as contemplated by the inventors.

Claims

1. A device for exchanging pressure between fluids, the device comprising: at least one tank comprising: a first side for receiving a first fluid at a higher pressure; and a second side for receiving a second fluid at a lower pressure; at least one piston in the at least one tank, the at least one piston configured to separate the first fluid from the second fluid, the at least one piston comprising a bypass feature, the bypass feature configured to: substantially prohibit the second fluid from traveling from the second side to the first side; and enable the first fluid to travel from the first side to the second side; a valve device at the first side of the at least one tank, the valve device configured to selectively place the first fluid at the higher pressure in communication with the second fluid at the lower pressure in order to pressurize the second fluid to a second higher pressure; and at least one sensor configured to detect a presence of the at least one piston, the at least one sensor spaced from an end of the at least one tank such that the at least one-piston is configured to travel past and beyond the at least one sensor toward the end of the at least one tank before contacting the end of the at least one tank.

2. The device of claim 1, wherein the at least one piston is independently disposed in the at least one tank and configured to travel unconstrained within the at least one tank from a first axial end to a second axial end of the at least one tank.
3. The device of claim 1, wherein the bypass feature of the at least one piston enables at least some of the first fluid and/or the second fluid to pass through the bypass feature in order to exit the at least one tank at the second side.
4. The device of claim 3, wherein the at least one piston is configured to approach and/or contact the second side of the at least one tank to force the at least some of the first fluid and/or the second fluid through the bypass feature of the at least one piston.
5. The device of claim 4, wherein the at least one piston approaching and/or contacting the second side of the at least one tank is configured to overcome a biasing force of the bypass feature to enable the at least some of the first fluid and/or the second fluid to travel through the bypass feature of the at least one piston.
6. The device of claim 1, wherein the bypass feature of the at least one piston enables the device to perform a fluid flush through the at least one tank from the first side to the second side.
7. The device of claim 1, wherein the bypass feature of the at least one piston is configured to at least partially prevent the second fluid from traveling through the bypass feature from the second side to the first side when the at least one piston is positioned proximate to the first side.
8. The device of claim 1, wherein the bypass feature of the at least one piston is configured to at least partially prevent the second fluid from traveling through the bypass feature to the valve device.
9. The device of claim 1, further comprising a controller configured to receive a signal from the at least one sensor on the at least one tank, the controller configured to, when the at least one sensor detects the at least one piston, instruct the valve device to increase flow of the first fluid at the higher pressure at the first side of the at least one tank in order to open the bypass feature in the at least one piston as the at least one piston approaches and/or contacts the second side.
10. The device of claim 9, wherein the controller is further configured to enable the at least one piston to contact the second side while at least partially preventing the at least one piston from contacting the first side.
11. The device of claim 1, further comprising a controller configured to receive a signal from the at least one sensor on the at least one tank, the controller configured to, when the at least one sensor detects the at least one piston, instruct the valve device to reduce flow of the first fluid at the higher pressure at the first side of the at least one tank in order to slow and stop movement of the at least one piston in the at least one tank as the at least one piston approaches and before the at least one piston contacts the first side or the second side.
12. A system for exchanging pressure between at least two fluid streams, the system comprising: a pressure exchange device for exchanging at least one property between fluids, the pressure exchange device comprising: at least one tank comprising: a first end for receiving a clean fluid with a first property; and a second end for receiving a dirty fluid with a second property; at least one piston in the at least one tank, the at least one piston configured to separate the clean fluid from the dirty fluid, wherein the at least one piston comprises at least one valve enabling fluid flow through the at least one piston; a valve device positioned at the first end of the at least one tank, the valve device configured to selectively place the clean fluid in communication with the dirty fluid in order to at least partially transfer the first property of the clean fluid to the dirty fluid, wherein the valve device is configured to force the at least one piston into contact with the second end of the at least one tank; and at least one sensor configured to detect a position of the at least one piston as the at least one piston passes the at least one sensor along a length of the at least one tank.
13. The system of claim 12, wherein the at least one valve in the at least one piston comprises a check valve.
14. The system of claim 13, wherein the check valve is configured to move to an open position

when a fluid traveling through the at least one piston overcomes a biasing force of the check valve.

15. The system of claim 14, wherein the check valve is configured to move to the open position to enable the clean fluid to pass to the second end of the at least one tank.

16. The system of claim 12, wherein the at least one sensor comprises at least two sensors configured to detect a presence of the at least one piston, wherein a first sensor is located near, but spaced from, the first end of the at least one tank and a second sensor is located near, but spaced from, the second end of the at least one tank.

17. The system of claim 16, wherein the at least two sensors are configured to each transmit a signal to the valve device responsive the position of the at least one piston and the valve device is configured to change from a first position to a second position responsive the signal from the first sensor and change from the second position to the first position responsive the signal from the second sensor.

18. A method of controlling a pressure exchange device comprising: supplying a high pressure fluid to a high pressure inlet of a valve configured to direct flow of the high pressure fluid to a chamber; transferring a pressure from the high pressure fluid to a dirty fluid in the chamber; monitoring a location of a piston with at least one sensor as the piston passes and travels beyond the at least one sensor along an axial length of the chamber; changing a position of the valve responsive the location of the piston; redirecting the flow of the high pressure fluid into the chamber by the changing of the position of the valve; and enabling some of the high pressure fluid to pass through the piston.

19. The method of claim 18, further comprising forcing the piston into a distal axial end of the chamber with the high pressure fluid to enable some of the high pressure fluid to pass through the piston, a proximate axial end of the chamber being positioned at the valve and the distal axial end being positioned away from the valve opposite the proximal axial end of the chamber.

20. The method of claim 18, further comprising substantially prohibiting the dirty fluid from passing through the piston to the high pressure fluid.
